Team 5

Phase-1 Report: Empirical Analysis of Catastrophic Forgetting in Continual Reinforcement Learning

Jeongbin Park, Lina Alkarmi, Henry Talbott

University of Michigan, Ann Arbor

Repo: github.com/JB243/ECE567_W2026_Team5_CRL

Project Track 2: Continual Reinforcement Learning

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1. Environment and Task Sequence

Minihack. The Minihack benchmark comprises 15 (train, test) navigation task pairs drawn from 27 NetHack based environments. Unlike Procgen, all tasks share the same visual assets making task-boundary detection ambiguous. It tests out-of-distribution generaliztion by testing on simpler variants (e.g. Room 5×5) and evaluating on harder ones (e.g. Room 15×15). Observations are 84×84 RGB frames with an 8-dim action space, run over 2 cycles.

Procgen. The Procgen benchmark uses six procedurally generated games: Climber, Dodgeball, Ninja, Starpilot, Bigfish, and Fruitbot. It tests in-distribution generalization. Agents train on 200 fixed levels and are evaluated on the full level distribution. Observations are 64×64 RGB frames (no framestacking) with a shared 15-dim action space. We train for 5M frames per task over 5 cycles (150M total).

Atari. The Atari benchmark consists of six Atari 2600 games: SpaceInvaders, Krull, BeamRider, Hero, StarGunner, and MsPacman. Tasks are visually distinct with little skill overlap, making it a test of catastrophic forgetting. Observations are 84×84 grayscale frames stacked over 4 timesteps with the full 18-dim action space.

CHORES. Finally, CHORES is a continual household robotics benchmark. It was created specifically for the CORA paper and depends on a custom demonstration dataset originally hosted by the authors. Unfortunately, the dataset link is no longer accessible and CHORES cannot be reproduced from the public codebase alone. We therefore omit this from our replication.

2. Baselines

All baselines build on the IMPALA actor-learner architecture and are implemented using the TorchBeast codebase. We evaluate five methods spanning three continual learning strategy categories:

IMPALA serves as the non-continual baseline with no mechanism to mitigate forgetting. **EWC** is a regularization method that penalizes changes to network weights deemed important for previous tasks, as estimated by the Fisher information matrix. **Online EWC** extends EWC by maintaining a running estimate of the Fisher information with exponential decay. **P&C** is an architectural method that alternates between a progress phase (active column adapts to new tasks) and a compress phase (knowledge distilled into a shared knowledge base). **CLEAR** is a rehearsal method that maintains an experience replay buffer using reservoir sampling, mixing past and present experience during training.

Not all baselines were evaluated on every benchmark. On MiniHack, we evaluate IMPALA and CLEAR only, consistent with the paper, using 5 seeds each. On Procgen, we evaluate all five methods (1 seed each; 5 cycles). We use the shallow IMPALA CNN rather than the deep residual model used in the paper, due to compute constraints (see Section 4). On Atari, we replicate CLEAR, P&C, and EWC.

3. Results

MiniHack: As shown in Figure 1, our results qualitatively match the paper: CLEAR outperforms IMPALA on early Room tasks, while both methods lose plasticity on later tasks (Corridor, River, HideNSeek). Confidence intervals are wider than the paper's due to fewer seeds.

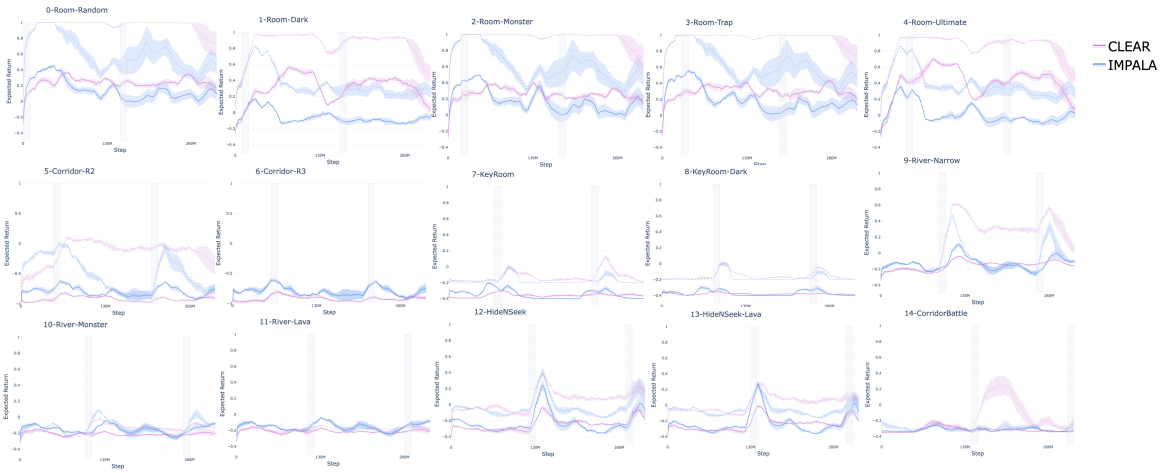

Figure 1: Minihack results.

Procgen: As shown in Figure 2, CLEAR outperforms other methods on most tasks, with non-CLEAR methods exhibiting catastrophic forgetting between training windows, and our results are comparable to those in the paper. However, on Bigfish and Fruitbot, trends are less clear, and on Climber, CLEAR's advantage is inconsistent. This is likely due to variance from a single seed.

Atari: As shown in Figure 3, CLEAR achieves the highest returns on most tasks, particularly SpaceInvaders, Hero, StarGunner, and MsPacman. P&C is competitive on Krull and BeamRider, while EWC consistently underperforms across all six games.

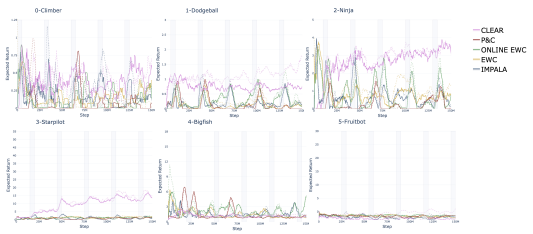

Figure 2: Procgen results.

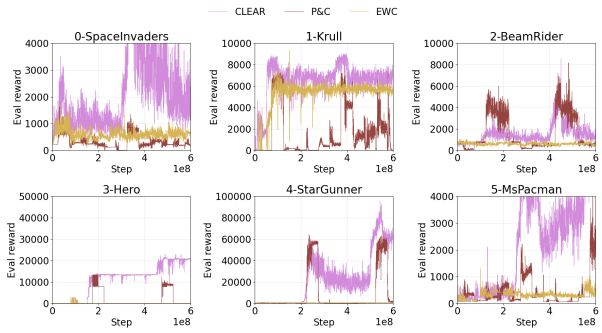

Figure 3: Atari results.

4. Reproducibility

Our replication matches the paper's qualitative trends across all benchmarks, despite several deviations due to compute constraints: we use 5 seeds for MiniHack (vs. 10), 1 seed per baseline for Procgen (vs. 20), and 1 seed per baseline (vs. 5). For Procgen, we use the shallow original IMPALA CNN rather than the paper's deep residual encoder. Overall, the reproduced benchmark trends are consistent with the original claims: replay-based continual RL remains a strong and practical baseline for retaining knowledge under substantial distribution shift.